INTERNAL — DISCOVERY REPORT

Management Summary

Live Discovery Report — azlocal-iic-01

Mode: current-state | Ranger Version: 2.1.0 | Generated: 04/06/2026 12:08:32

- Overall Health
- Azure Integration
- Security Posture
- Monitoring Coverage

Storage used:	0%
Storage reserve:	0%
Safe allocatable:	0%
Monitoring coverage:	100%
Backup coverage:	20%

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Priority Recommendations

- **[WARNING]** One or more node certificates expire within 90 days
  Review certificate ownership and renew expiring node certificates before handoff.
- **[WARNING]** iDRAC firmware below recommended baseline on all nodes
  Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.
- **[INFORMATIONAL]** Azure Policy assignments discovered at resource group scope
  Verify policy scope extends to arc resource group and monitor for compliance drift.
- **[INFORMATIONAL]** Cluster event history includes a transient network quorum warning
  Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.

Collector Overview

hardware

Status: success

Targets:

management-performance

Status: success

Targets:

monitoring-observability

Status: success

Targets:

storage-networking

Status: success

Targets:

topology-cluster

Status: success

Targets:

waf-assessment

Status: success

Targets:

workload-identity-azure

Status: success

Targets:

Run Summary

- Mode: current-state
- Generated: 04/06/2026 12:08:32
- Cluster: azlocal-iic-01
- Nodes: 3
- Collectors: 7/7 successful
- Artifacts currently recorded: 0

Readiness Snapshot

- Schema validation: passed
- Critical findings: 0
- Warning findings: 2
- Informational findings: 2
- Successful collectors: 7 of 7
- Partial or failed collectors: 0

Health Status

- Overall health: GREEN (critical, warning findings)
- Azure integration: GREEN (10 Azure resources discovered)
- Security posture: GRAY (Secured-Core enabled on node(s))
- Monitoring coverage: GREEN (100% of nodes have Azure Monitor Agent)

Environment Overview

- VMs discovered: 5
- Azure resources discovered: 10
- Connected Azure auth method: service-principal
- Monitoring resources: 5
- Management services running: 4

Topology and Operating Model

- Deployment type: hyperconverged
- Identity mode: ad
- Connectivity mode: connected
- Storage architecture: storage-spaces-direct
- Network architecture: switched
- Variant markers: connected

Domain Coverage

- Cluster nodes: 3
- Cluster roles: 5
- Storage pools: 1
- Physical disks: 24
- Network adapters: 12
- VMs discovered: 5
- Azure resources: 10
- Alerting resources: 2
- Management services: 4

Node Inventory

Node	Model	State	OS	OS Version / Build	Logical CPUs	RAM (GiB)
azl-iic-n01	PowerEdge R760	Up	Microsoft Azure Stack HCI	10.0.25398.1189	64	512
azl-iic-n02	PowerEdge R760	Up	Microsoft Azure Stack HCI	10.0.25398.1189	64	512
azl-iic-n03	PowerEdge R760	Up	Microsoft Azure Stack HCI	10.0.25398.1189	64	512

Collector Status

- hardware: success ()
- management-performance: success ()
- monitoring-observability: success ()
- storage-networking: success ()
- topology-cluster: success ()
- waf-assessment: success ()
- workload-identity-azure: success ()

Operational Risk Summary

- Nodes not healthy: 0
- Unhealthy disks: 0
- High CPU nodes: 0
- Certificates expiring within 90 days: 1
- Azure Policy assignments: 2
- Schema warnings: 0
- Azure Advisor recommendations: total, 3 high-impact

Workload Summary

- Total VMs: 5
- Total nodes: 3
- AKS clusters: 1
- Arc-connected machines: 1
- Update compliance: No update resources tracked
- Licensing: IIC Platform Production
- VMs using Arc IP fallback: Not collected

Capacity Summary

- Storage total raw: 0 TiB
- Storage total usable: 0 TiB
- Storage used by workloads: 0 TiB (0% of usable)
- Storage reserve target: 0 TiB (0% of usable)
- Storage free usable: 0 TiB
- Projected safe allocatable: 0 TiB
- Thin provisioning ratio: Not computed
- vCPU:pCPU overcommit ratio: Not computed
- Memory overcommit ratio: Not computed
- Average VMs per node: N/A

Priority Recommendations

- [WARNING] One or more node certificates expire within 90 days: Review certificate ownership and renew expiring node certificates before handoff.
- [WARNING] iDRAC firmware below recommended baseline on all nodes: Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.
- [INFORMATIONAL] Azure Policy assignments discovered at resource group scope: Verify policy scope extends to arc resource group and monitor for compliance drift.
- [INFORMATIONAL] Cluster event history includes a transient network quorum warning: Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.

VM Inventory

Name	State	vCPU	RAM (GiB)	Host Node	Generation
avd-iic-sh01	Running	8	16	azl-iic-n01	2
avd-iic-sh02	Running	8	16	azl-iic-n02	2
avd-iic-sh03	Running	8	16	azl-iic-n03	2
arc-iic-vm01	Running	4	8	azl-iic-n01	2
arc-iic-vm02	Running	4	8	azl-iic-n02	2

VM Density Metrics

- VMs per node (average): N/A
- Highest-density node: N/A
- vCPU:pCPU overcommit ratio: Not computed
- Memory overcommit ratio: Not computed
- Arc-connected VMs:
- VMs using Arc IP fallback:
- Avg CPU utilization (all nodes): 38.3%
- Avg available memory (all nodes): 176 GiB

Storage Pool Capacity

Pool	Raw (GiB)	Usable (GiB)	Used (GiB)	Reserve (GiB)	Safe Alloc (GiB)	Posture
—	0	0	0	0	0	—

ESU Enrollment

- Eligible Arc-connected VMs:
- Enrolled in ESU:
- Not enrolled in ESU:
- Ineligible:
-

Coverage Assessment

- Monitoring coverage (AMA): 100% (3 of 3 nodes have Azure Monitor Agent)
- Backup coverage estimate: 20% (based on backup items vs VM count)
- Defender for Cloud: Not confirmed or not collected
- WDAC policy: Not collected
- BitLocker status: Not collected
- Certificates expiring in <90 days: 1

Security Posture Summary

- Secured-Core enabled nodes: of 3
- Syslog forwarding nodes:
- RBAC assignments at resource group: 1
- Policy assignments tracked: 2
- Policy exemptions:
- Active Directory site: Not collected

Management Tool Coverage

- WAC installed nodes: of 3
- SCVMM agent nodes:
- SCOM agent nodes:
- Running management services: 4
- Third-party agent types: (0)

Cost & Licensing

- AHB status: Enabled
- Total physical cores: 96 | AHB-covered: 96 | Unenrolled: 0
- Current monthly cost: 960 USD @ 10/core/month
- AHB adoption: 100%
- Potential monthly savings (if remaining cores enrolled in AHB): 0 USD
- Pricing based on Azure Local public pricing (10 USD/physical core/month) as of 2026-04-16.
- For current rates, see: https://azure.microsoft.com/en-us/pricing/details/azure-local/

Cost & Licensing — Per Node

Node	Physical Cores	AHB Enabled	Monthly Cost (USD)	Monthly Saving (USD)
azl-iic-n01	32	Yes	320	320
azl-iic-n02	32	Yes	320	320
azl-iic-n03	32	Yes	320	320

Pricing as of 2026-04-16 — https://azure.microsoft.com/en-us/pricing/details/azure-local/

Arc Extensions by Node

Node	Name	Type	Publisher	Version	State
azl-iic-n01	AzureMonitorWindowsAgent	AzureMonitorWindowsAgent	Microsoft.Azure.Monitor	1.22.2.0	Succeeded
azl-iic-n01	GuestConfigExtension	ConfigurationforWindows	Microsoft.GuestConfiguration	1.29.78.0	Succeeded
azl-iic-n01	MicrosoftDefenderForServers	MDE.Windows	Microsoft.Azure.AzureDefenderForServers	1.0	Succeeded
azl-iic-n02	AzureMonitorWindowsAgent	AzureMonitorWindowsAgent	Microsoft.Azure.Monitor	1.22.2.0	Succeeded
azl-iic-n02	GuestConfigExtension	ConfigurationforWindows	Microsoft.GuestConfiguration	1.29.78.0	Succeeded
azl-iic-n02	MicrosoftDefenderForServers	MDE.Windows	Microsoft.Azure.AzureDefenderForServers	1.0	Succeeded
azl-iic-n03	AzureMonitorWindowsAgent	AzureMonitorWindowsAgent	Microsoft.Azure.Monitor	1.22.2.0	Succeeded
azl-iic-n03	GuestConfigExtension	ConfigurationforWindows	Microsoft.GuestConfiguration	1.29.78.0	Succeeded
azl-iic-n03	MicrosoftDefenderForServers	MDE.Windows	Microsoft.Azure.AzureDefenderForServers	1.0	Succeeded

AMA coverage: 100% of nodes. Failed extensions: 0.

Logical Networks

Name	VM Switch	DHCP	Subnets	State
lnet-iic-mgmt-01	ConvergedSwitch	No	1	Succeeded
lnet-iic-workload-01	ConvergedSwitch	Yes	1	Succeeded

Logical Network Subnets

Network	Subnet	Address Prefix	VLAN	IP Pools
lnet-iic-mgmt-01	subnet-management	10.0.0.0/24	2203	1
lnet-iic-workload-01	subnet-workload	10.0.2.0/24	2205	1

Storage Paths

Name	Path	Available (GB)	File System	State
sp-iic-vmstore-01	C:\ClusterStorage\csv-iic-azlocal-01-vmstore-01\ArcVmDisks	1480	CSVFS_ReFS	Succeeded
sp-iic-vmstore-02	C:\ClusterStorage\csv-iic-azlocal-01-vmstore-02\ArcVmDisks	1620	CSVFS_ReFS	Succeeded
sp-iic-vmstore-03	C:\ClusterStorage\csv-iic-azlocal-01-vmstore-03\ArcVmDisks	1710	CSVFS_ReFS	Succeeded

Arc Resource Bridge

Name	Status	Version	Distro	Provisioning
rb-iic-azlocal-01				
Connected				
1.5.23				
MOC-K8S				
Succeeded				

Custom Locations

Name	Namespace	Location	State
cl-iic-azlocal-01			
azlocal			
eastus			
Succeeded			

Arc Gateways

Name	Endpoint	Allowed Features	State
arcgw-iic-01			
https://arcgw-iic-01.gw.arc.azure.com			
*			
Succeeded			

Marketplace & Custom Images

Name	Type	OS	Version	Size (GB)	State
img-iic-ws2022-datacenter	Marketplace	Windows	20348.2700.240906	127	Succeeded
img-iic-rhel9-base	Custom	Linux	9.4.1	24	Succeeded

Arc Agent Versions

Version	Node Count	Nodes
1.39.02750.1478	2	azl-iic-n01, azl-iic-n02
1.38.02620.1234	1	azl-iic-n03

Drift status: warning. Latest: 1.39.02750.1478. Max behind: 1.

VM Distribution by Node

Node	VM Count
azl-iic-n01	2
azl-iic-n02	2
azl-iic-n03	1

Balance: balanced | CV: 0.283 | Status: warning

WAF Assessment — Scorecard

Pillar	Score	Status	Rules Passing	Top Finding
Reliability	88%	Excellent	8 / 9	No unhealthy storage pools
Security	44%	Fair	2 / 8	Secured-Core is enabled on cluster nodes
Cost Optimization	91%	Excellent	2 / 3	ESU-eligible VMs are enrolled in Extended Security Updates
Operational Excellence	50%	Fair	5 / 9	Azure Monitor alert rules are defined
Performance Efficiency	60%	Good	2 / 3	Storage utilization is within safe limits

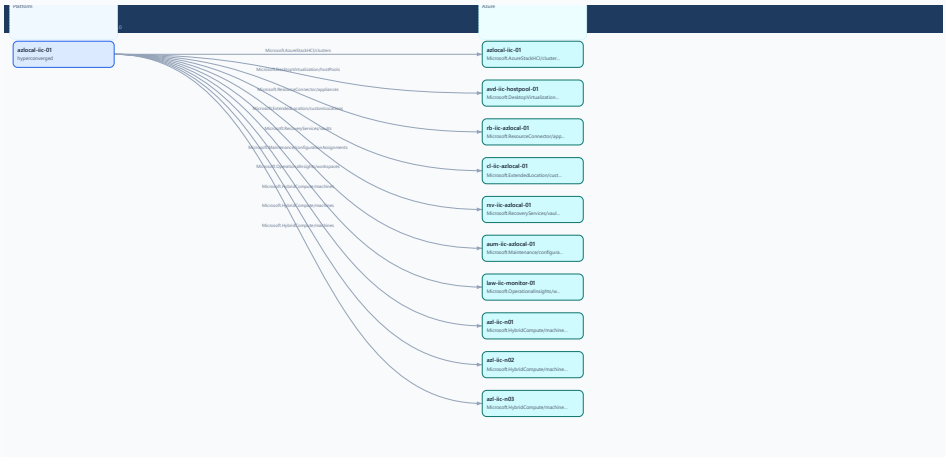
Overall WAF score: 67% (19 of 32 rules passing). Evaluated from saved manifest — no re-collection required.

WAF Assessment — Azure Advisor Recommendations

Pillar	Impact	Finding	Recommendation
—	High	System.Collections.Specialized.OrderedDictionary	—
—	High	System.Collections.Specialized.OrderedDictionary	—
—	High	System.Collections.Specialized.OrderedDictionary	—
—	Medium	System.Collections.Specialized.OrderedDictionary	—
—	Medium	System.Collections.Specialized.OrderedDictionary	—
—	Medium	System.Collections.Specialized.OrderedDictionary	—
—	Medium	System.Collections.Specialized.OrderedDictionary	—
—	Medium	System.Collections.Specialized.OrderedDictionary	—

Architecture Diagrams

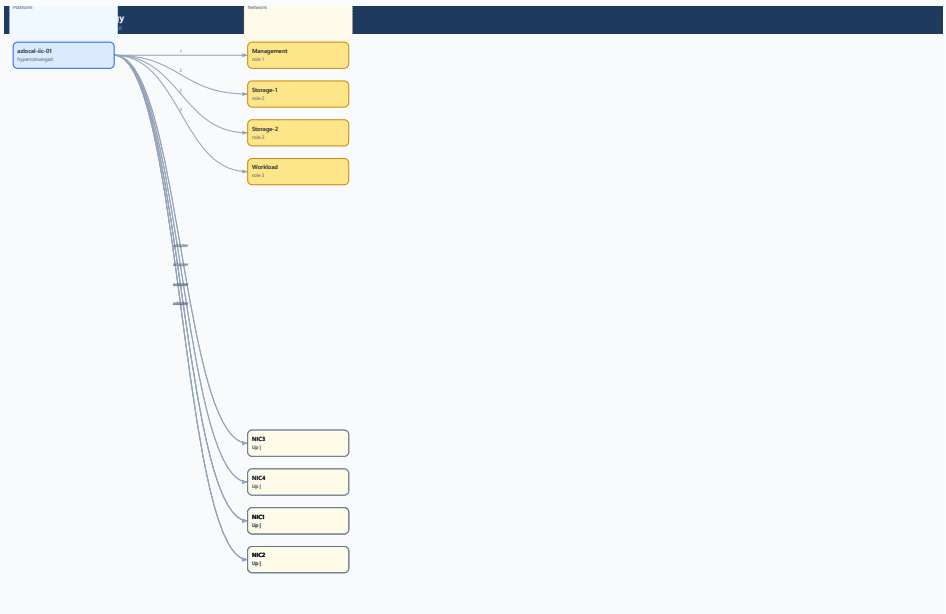
Azure Arc Integration



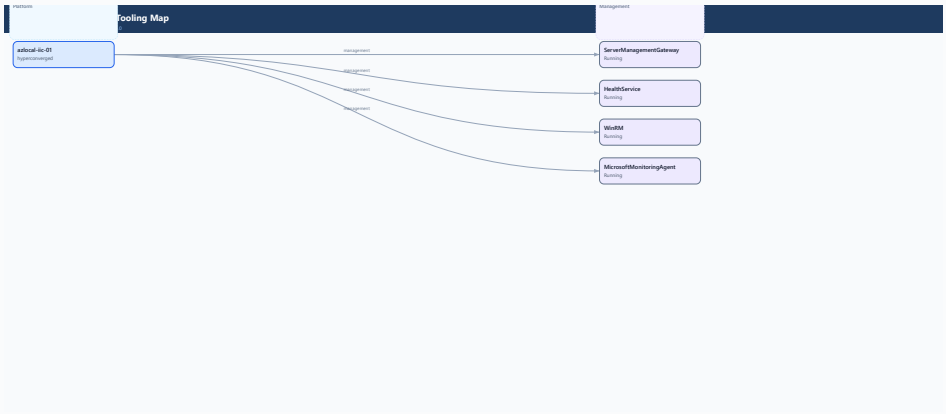
Backup Recovery Map

Identity Access Surface

Logical Network Topology

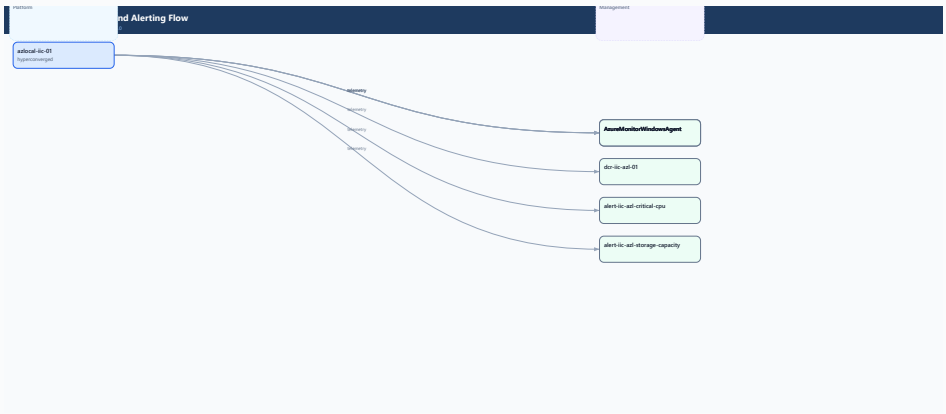


Management Plane Tooling



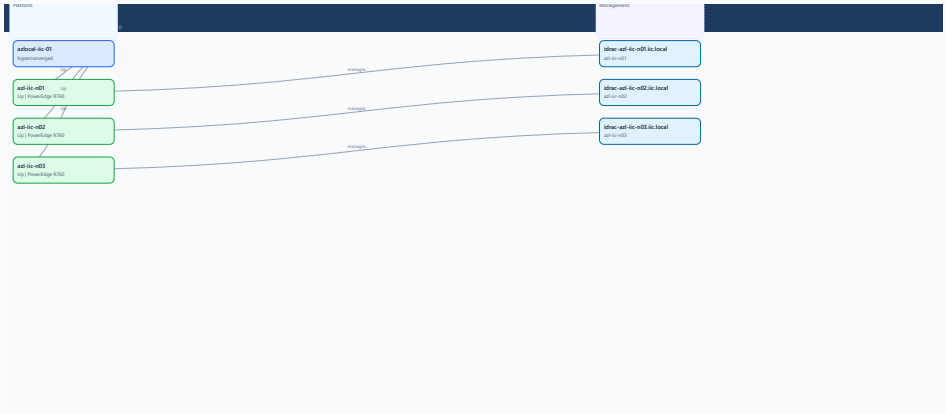
Monitoring Health Heatmap

Monitoring Telemetry Flow



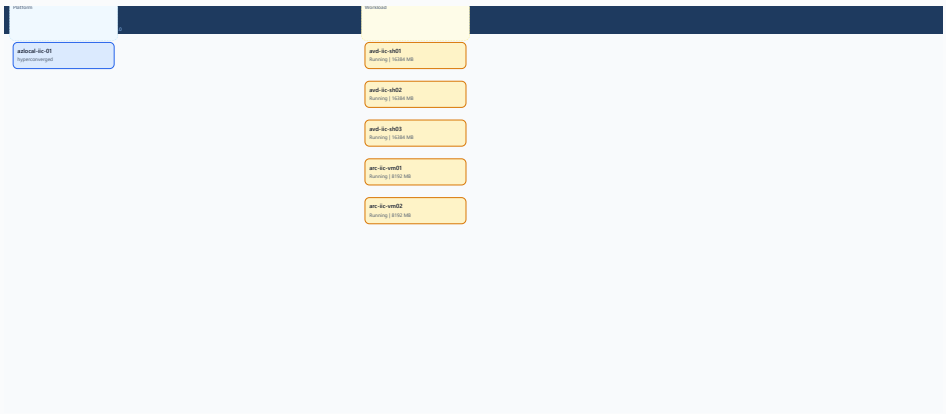
Oem Firmware Posture

Physical Architecture

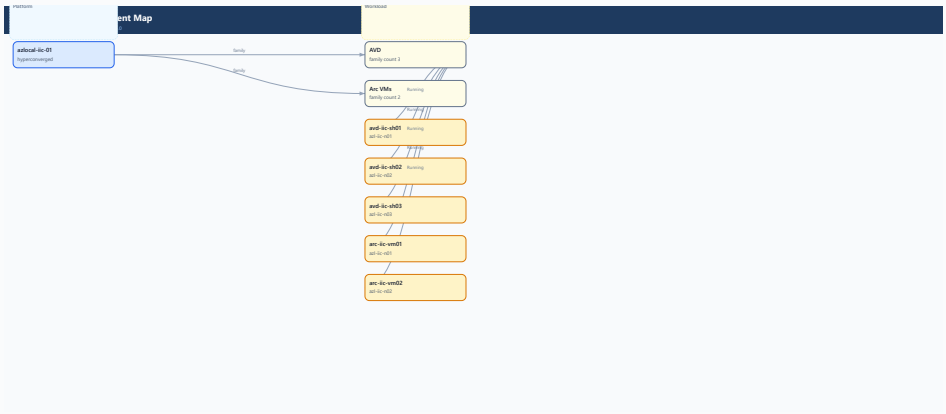


Storage Architecture

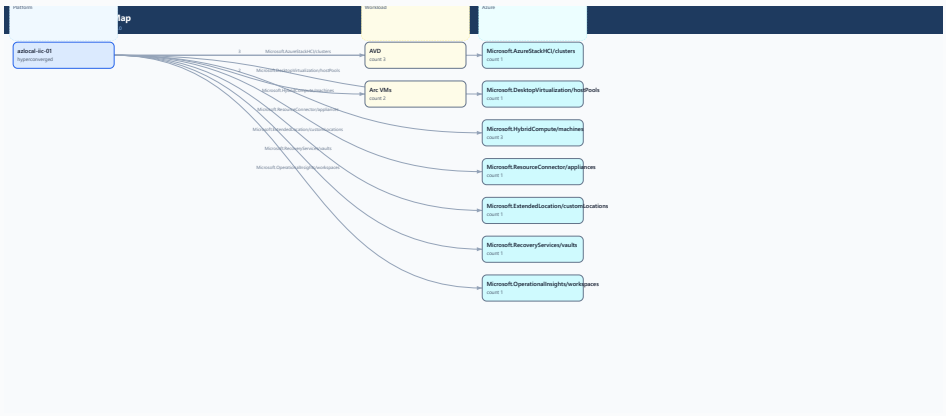
Vm Placement Map



Workload Family Placement



Workload Services Map



Findings

WARNING **One or more node certificates expire within 90 days**

Identity posture data indicates certificate on azl-iic-n01 expires within 90 days.

Current state: 1 certificate expiring within 90 days

Recommendation: Review certificate ownership and renew expiring node certificates before handoff.

WARNING **iDRAC firmware below recommended baseline on all nodes**

OEM integration data shows iDRAC firmware at 7.10.30.00. The recommended baseline is 7.20 or later.

Current state: iDRAC firmware 7.10.30.00

Recommendation: Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.

INFORMATIONAL **Azure Policy assignments discovered at resource group scope**

2 policy assignments are enforcing tagging and security control auditing.

Current state: 2 policy assignments active

Recommendation: Verify policy scope extends to arc resource group and monitor for compliance drift.

INFORMATIONAL **Cluster event history includes a transient network quorum warning**

A Management network quorum warning event was recorded on 2026-04-06. Cluster recovered automatically.

Current state: single event logged; cluster healthy

Recommendation: Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.